

Lab Manual

The Physics of Climate Change

Benjamin Holder
GVSU Physics
Fall 2025

Contents

1	Measurement, Uncertainty, and Temperature	5
1.1	Introduction and Definitions	5
1.2	Experiments	7
1.2.1	The time of my name	7
1.2.2	“Seeing” the uncertainty I: Room Temperature	7
1.2.3	“Seeing” the uncertainty II: Body Temperature	8
1.2.4	Leslie Cube	9
1.2.5	Temperatures of other objects	10
2	Global Average Temperature	13
2.1	Introduction and Definitions	13
2.2	Experiments and Activities	14
2.2.1	Temperature Anomalies	14
2.2.2	The approach to an average	15
2.2.3	Global average temperature (of your table)	15
2.2.4	Global average temperature (of the Earth land surface)	16
3	Energy and Equilibrium	17
3.1	Introduction and Definitions	17
3.1.1	Motion	17
3.1.2	Dynamics: Newton’s Law(s)	18
3.1.3	Energy: Mechanical Energy	19
3.1.4	Energy: Other Energy	19
3.1.5	Energy: Equilibrium	19
3.2	Experiments and Activities	20
3.2.1	Motion	20
3.2.2	Mechanical Work	20
3.2.3	Work and thermal energy	20
3.2.4	Equilibrium: Leaky Buckets	21
4	Electromagnetic Radiation and Blackbody Radiation	25
4.1	Introduction and Definitions	25
4.1.1	Electromagnetic Spectrum	26
4.1.2	Blackbody Radiation	27

4.2	Experiments and Activities	28
4.2.1	Observations of light	28
4.2.2	Observations of transmission of light	29
5	Heating by Radiation	31
5.1	Introduction and Definitions	31
5.1.1	First Law of Thermodynamics, Heating, and Calorimetry	32
5.2	Experiments and Activities	34
5.2.1	Specific heat of aluminum	34
5.2.2	Heating aluminum block by radiation	35

Lab 1

Measurement, Uncertainty, and Temperature

Say my name. Say my name.

Equipment List

- Leslie Cube
- Glass Thermometer
- IR Thermometer “gun” (2)
- Stopwatch
- Ruler
- Computer

Overview

In this weekly lab you will primarily work in pairs, with your “partner” at the same table, but feel free to discuss things with the “group” of four at your adjoining tables. We will also share data with the entire class, usually using google surveys and sheets. You will turn in your own individual lab assignment, with copies of all data and graphs, in a single pdf document uploaded to the Blackboard “assignment” for that week.

1.1 Introduction and Definitions

Today we will conduct our first measurements of physical phenomena, mostly temperature measurements, and understand the necessity of the uncertainty and units associated with those measurements.

A **physical measurement** contains three equally important parts: the numerical **value**, the **uncertainty** (also referred to as *error*) in that value, and the **units** in which these are measured.

Some examples are:

- The human body temperature is 37.0 ± 0.5 °C.
(Uncertainty represented as “plus or minus” makes a range of values. Both the value and uncertainty are in “degrees celsius”.)
- A college basketball game has 40 minutes ± 0.1 seconds of playing time.
(The final minute clock measures to tenths of a second. The units can be different between the value and uncertainty.)
- The Sun is 93 million miles away
Here the uncertainty is expressed *implicitly* as “plus or minus a few of the last digit given”, so plus or minus a few million miles

Let's make a precise definition of the uncertainty/error in a measurement:

Given a particular measurement, the **uncertainty** is a statement of the range of values in which another scientist is likely¹ to make the same measurement.

There are a number of sources/components of this uncertainty:

- **Instrumental Uncertainty:** A measurement device generally has a smallest unit division shown. For an analog device — like a ruler with millimeter marks, or a thermometer with degrees — where we might hope to measure between the smallest markings, we can specify the instrumental uncertainty as such (e.g., half of a millimeter on a ruler, or 0.3°C on a thermometer). For a digital device, we can say that the instrumental uncertainty is “plus or minus a few of the last digit” (e.g., ± 0.03 seconds on a standard stopwatch).
- **(Manufacturing) Instrumental Uncertainty:** Sometimes although an instrument shows a precise demarcation, it is clear from looking at multiple “identical” instruments that they were not manufactured to that precision (e.g., multiple identical thermometers read slightly different values). The range of manufacturing uncertainty can be assessed by using multiple copies of the same device.
- **Methodological Uncertainty:** Sometimes our uncertainty is limited more by the way we make a measurement than by the instrument itself. For example, although a stopwatch has instrumental uncertainty in the hundredths of a second, our human reaction time is no better than a few *tenths* of a second, so our methodological uncertainty would be ± 0.3 s. This can be assessed by making repeated measurements.
- **Natural Variation** Almost always, the object of measurement does not have an infinitely precise value (e.g., the “temperature outside right now” depends on where exactly you measure it, and at which moment in time; the “diameter of a rock” that is not spherical). This uncertainty can sometimes be limited by a more precise specification of the measurement (e.g., “the temperature in the shade, outside Padnos, at 10:40 am”; or “the largest diameter of the rock”). And, like the methodological uncertainty, it can be assessed by repeated measurements.

¹More precisely, if we say that the uncertainty is the *standard deviation* over repeated measurements, then 68% of all measurements fall in that range

1.2 Experiments

1.2.1 The time of my name

- Have your partner recite their full name to you at a steady relaxed pace. Measure the amount of time this takes using the stopwatch. You can try this multiple times. Then switch and recite your own name for them.

Questions

1. Write down the name, and then your time measurement with uncertainty and units.
2. Briefly explain how you arrived at the uncertainty value, making mention of all “sources” of uncertainty.

1.2.2 “Seeing” the uncertainty I: Room Temperature

One table has been set up with an array of thermometers. Since they are all localized in the same place, they should be making approximately the same measurement: the temperature of the air at that table right now.

- Take a careful clear photo of the array of thermometers, then return to your desk and write down each of the temperature measurements on a sheet of paper.
- Considering the range of measurements you have, put labeled tick marks on the axis below such that the values will be spread out across the line. Use a ruler to make equal spacing of tick marks. At the far right, label that axis (e.g., “Temperature (C)”).
- Place a dot on/near the axis for every measurement of temperature. You can “scatter” these a bit above/below the axis for clarity.



Questions

1. Express the full measurement with value, uncertainty, units. Estimate the uncertainty using your graphical display of points.
2. What is the instrumental uncertainty for this thermometer?
3. What do you think is the primary source of the uncertainty? Mention each type of uncertainty source.

1.2.3 “Seeing” the uncertainty II: Body Temperature

Here each of you will make a measurement of your body temperature using the Infrared (IR) thermometer “gun” on the palm of your hand. Note the picture on the side of the IR thermometer gun describing its “beam” of measurement (imagine a spot being projected onto the thing you are measuring, and make sure the spot is contained within the object). You will produce a two-picture plot, using a provided code (see Blackboard) and the Google CoLab implementation of python programming language.

1. What is the instrumental uncertainty associated to the IR thermometer?

2. Do you observe any natural variation of temperature across your hand?
 3. Write down your measurement (with value, uncertainty, and units) here, and then submit it to the google survey (see Blackboard).
- Copy the column of “values” data for the whole class in the linked spreadsheet (see Blackboard) into your own spreadsheet and save as a csv file.
 - In a web browser, go to `colab.research.google.com` and open a “New Notebook”.
 - Paste in the code provided on blackboard for *Scatter and Histogram Plot*.
 - To run the code, either press the “play” button, or press shift-enter/shift-return. It will prompt you for your csv file to upload.
 - This will produce a plot on the screen.
 - Notice that there are generic axis labels and titles. Find the place in the code where they have been entered and edit them to make them specific to your particular data.
 - To save and print your plot, you can either right-click and save to a file or screenshot to a file. Print this plot so that you can add it to your submission.

1.2.4 Leslie Cube

- Fill up the Leslie cube with hot/warm water at the sink and then replace the rubber stopper. Fill a small beaker with cold water.
 - Place the glass thermometer first in the cold water, and then insert it into the Leslie cube such that its bulb is in the middle of the cube.
1. Record the glass thermometer measurements of the temperature in the cold water and in the Leslie cube.
 2. Describe how quickly the thermometer came to equilibrium with the cold water and with the hot water.
 3. Take and record temperature measurements of each side of the Leslie cube using the IR thermometer gun.

1.2.5 Temperatures of other objects

- Using the IR thermometer gun, record measurements of temperature (with value, uncertainty, units for each) for some of the following objects:
 - The surface of the table you sitting at.
 - Something else in the room.
 - Outside: Something that might be in equilibrium with the air temperature (shaded spot).
 - Outside: Something that is probably not in equilibrium with the air temperature.
 - Outside: The sky (try both clear and clouded spots).
- Record your temperature measurements below and also report them to the google survey posted on Blackboard.

1. Record your temperature measurements here.

Summary Questions

Brief answers are fine, but please write in complete sentences.

1. State the temperature measurements you made for the Leslie cube, including each type of side and the direct glass thermometer measurement. Speculate on why you found different values, if you did.

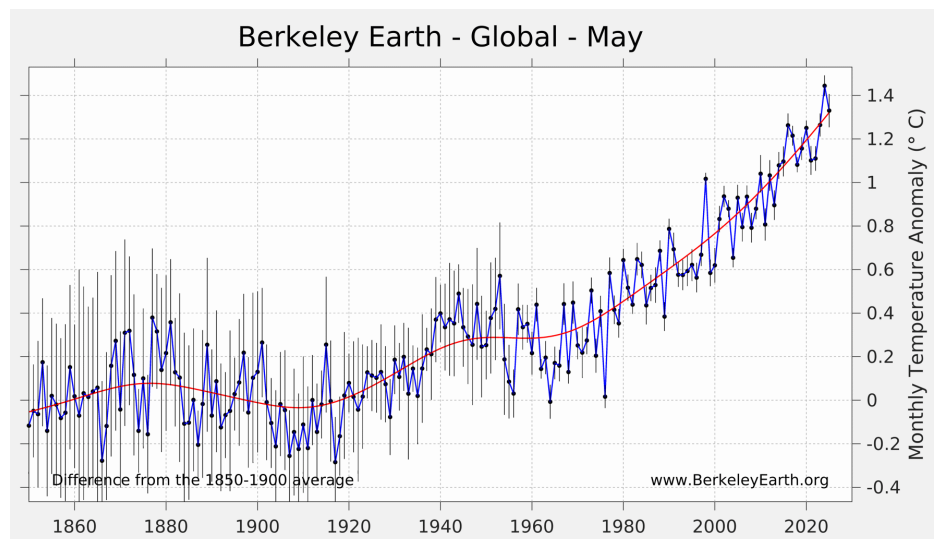
2. Looking at either the hand-temperature data or the other plot you made with the entire class data: how did your own estimated uncertainty (for your measured value) compare to the “spread” of the data taken by all students? Should these agree? Why or why not?

3. Two students make a measurement of the distance from the Bookstore to the Clock Tower. They are reported as follows

$$199.65 \text{ m} \pm 0.3 \text{ m}$$

$$202 \text{ m} \pm 3 \text{ m}$$

- (i) On the axis below, draw tick marks over a wide enough range to plot both measurements *and their ranges of uncertainty*. (ii) Plot the two measurements using a point for the “value” and a horizontal bar to show the uncertainty. (iii) Are these “the same” measurements to within uncertainty or not? Explain.
4. Shown below is a summary of the Global Average Temperature over most of human recorded history. The uncertainties for each data point are indicated by bars. Speculate on how these uncertainties might have been determined, how they change over time, and why.



Lab 2

Global Average Temperature

The anomaly is never abnormal. It's not even anomalous.

Equipment List

- Ruler
- Meter Stick
- IR Thermometer “gun”
- Masking Tape
- incandescent lamp
- Computer

Note: When you arrive at your table, clear a space on your table that is approximately half the area and mark it off with masking tape. Then turn on the lamp so that it is pointing down at one portion of that empty space. You are “pre-heating” the table for measurements we will make below.

2.1 Introduction and Definitions

Today we consider the *Global Average Temperature*, which we have seen is the primary measure used to specify the degree of global climate change over time. Specifically, we will perform some experiments and analyses to mimic the process by which this quantity is calculated. Obviously, there are many details that are used by professional climatologists that we will ignore, but we will focus on the two most important ideas: *area-weighted average* and *temperature anomalies*.

The **global average temperature** at a given moment in time, is defined to be (and calculated as) the weighted average of the temperature at every point on Earth, with weight for a given point being the area fraction assigned to that point. For example, if we restricted ourselves to just the United States, we could *estimate* the “US average temperature” as:

$$T_{\text{US,avg}} = T_{\text{MI}} \frac{A_{\text{MI}}}{A_{\text{USA}}} + T_{\text{IN}} \frac{A_{\text{IN}}}{A_{\text{USA}}} + T_{\text{IL}} \frac{A_{\text{IL}}}{A_{\text{USA}}} + \dots$$

where you can see that the temperature value assigned to a state is weighted by the fraction of area (measured, e.g., in square kilometers) that state has in the total area of the US. This can

be written equivalently as

$$T_{\text{US,avg}} = \frac{T_{\text{MI}} A_{\text{MI}} + T_{\text{IN}} A_{\text{IN}} + T_{\text{IL}} A_{\text{IL}} + \dots}{A_{\text{USA}}}$$

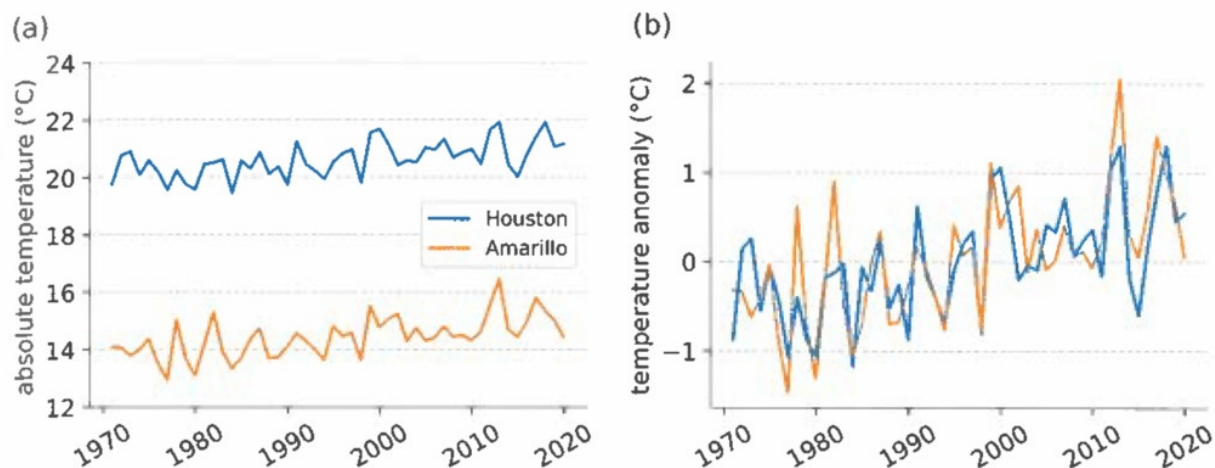
This is a *weighted average* because the sum of all the factors in the numerator, $A_{\text{MI}} + A_{\text{IN}} + A_{\text{IL}} + \dots$, adds up exactly to the value in the denominator (or, in the first question, the sum of all the weights, e.g., $\frac{A_{\text{MI}}}{A_{\text{USA}}}$, add up to one). The calculation above would only be an estimate of the average temperature, because we have used only 50 points (one for each state) instead of “all” points in the US. To improve our estimate¹, we could use multiple points per state (perhaps using a thermometer in each major city). Today we will perform a number of calculations to demonstrate this technique.

However, it is not actually the average temperature that is used to track climate change, but what is called the **global average temperature anomaly**. The average temperature anomaly is calculated as follows. (1) A *reference time range* is chosen (usually a span of a decade or more, e.g., 1950–1980), over which temperatures will be averaged to determine a (local) reference temperature value. (2) At each thermometer location, the *local reference temperature* is calculated as an average over the reference time range. (3) The temperature anomaly (over time) at a given thermometer is calculated by subtracting off the local reference temperature from all temperature values. (4) An area-weighted average of the temperature anomalies is calculated over the entire Earth.

2.2 Experiments and Activities

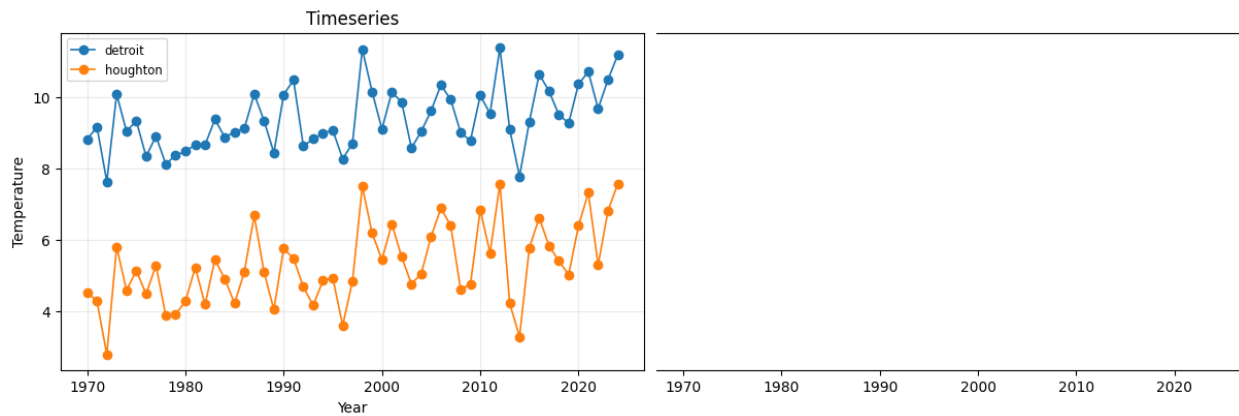
2.2.1 Temperature Anomalies

Recall Figure 2.1 from the Dessler textbook, showing the temperature over time for two Texas cities, and then their respective temperature anomalies:



¹Technically: when the global average temperature is calculated, the discrete temperature measurements at weather stations are used to produce a continuous “function” of temperature at every point on earth by interpolation between the stations; then that continuous function is “integrated” (calculus) over the total area.

Use the similar graph below, for two Michigan cities. (i) Determine a reference time period (ii) Estimate the reference temperature for each (iii) Make an anomaly temperature scale at right, and plot the values at three years for each city.



2.2.2 The approach to an average

Find the dataset of your own collected temperature measurements (submitted by you and your classmates to the google form this past week) in the “Results” link on Blackboard. For each of the two types of temperatures (air and sky) perform the following analysis:

- Copy the data from the spreadsheet of all data into your own Google sheet (or Excel spreadsheet).
- Clean the data column to make sure it contains only one numerical value per row. Save the single column of cleaned data as a csv file.
- Find the “Scatter-Histogram-Average” code on Blackboard. Copy the blocks of code into a new Google Colab notebook. (This step does not need to be done twice).
- Run the code blocks. The last block will prompt you to upload your csv file and then make a plot.

2.2.3 Global average temperature (of your table)

Perform a sequence of area-weighted averages of the temperature of your table in the previously marked off region, by completing the following steps:

1. Measure the total area that you have marked off and record this value.
2. Choose one point in the total area and measure the temperature using your IR thermometer gun. This is **Estimate 1**.
3. Using masking tape, partition your total area into 3–4 smaller areas. Measure the area of each one and record their values. Also draw a picture to show the way you have partitioned and to give names to the sub-areas (e.g., A_1 , A_2 , etc).
4. Measure and record the temperature at one point in each of the sub-areas. Create a table with columns [name, area (units), Temperature (units)].
5. Using the method described above (average US temperature), calculate an area-weighted temperature from your data. This is **Estimate 2**.

6. Subdivide your table even further, such that you have about twice as many sub-areas. Repeat steps 3–5 to calculate **Estimate 3**.

2.2.4 Global average temperature (of the Earth land surface)

We will now attempt to find the global average temperature anomaly for the Earth. Find the “Get annual temperature data for one location” code on Blackboard, and import the code blocks into a new Google Colab notebook. We will make three estimates:

- **Estimate 1:** Using only one location on Earth (your choice).
- **Estimate 2:** Using one location on each continent (again, your choice per continent).
- **Estimate 3:** Using about twice as many locations, spread out across the Earth.

In each trial you will take the following steps:

1. Use the “Get annual temperature data” code to download an annual average temperature dataset for one or more location over the 1970–2024 time range.
2. Choose a time range over which to determine the local reference average (e.g., 1970–1990).
3. Open this csv file (in Google sheets or Excel), and highlight the cells in that reference time range to find the reference average value.
4. Use the spreadsheet tools to make a new temperature anomaly column, in which the reference average is subtracted off from the temperature.
5. In Estimates 2 and 3, where you have multiple locations, create a new spreadsheet with a single date column at left and then a temperature anomaly column for each location.

To make the global average calculation, we will need to edit this file in a few additional steps:

- Look up online the surface area of each continent in square kilometers and record them. To make these values more understandable, let’s convert each into units of “Michigan mitten”. The lower peninsula of Michigan has an area of almost exactly 100,000 square kilometers. So you can divide each continental area by that value to obtain the area in “mittens”.
- In your multiple location spreadsheet, add two blank rows at top. For each anomaly column, put the name in the first row, and the area (in mittens) in the second row. In the first column (of dates) you can leave the first two rows blank.
- Locate the “Get weighted average of timeseries” code on blackboard and import the blocks into Colab. It will allow you to import your csv file. If you use the format prescribed above (two extra rows above the temperature vs time data) then it should produced a weighted temperature anomaly graph.

Lab 3

Energy and Equilibrium

There's a hole in the bucket, dear Liza, dear Liza

Equipment List

- Ruler
- Stopwatch
- Graph paper
- Glass thermometer, plastic rod stirrer (or some way of shaking?), 50mL grad cyl (2?)
- Leaky Bucket Kit (pitcher, 100 mL grad cyln, 25 mL grad cyl, aquarium pump, pump power control, two tubes, two binder clips)
- Track, cars, track-pulley, mass hanger, string (in the room, to be set up by students)

3.1 Introduction and Definitions

Today we first introduce Newton's dynamical theory of *motion*, for the purpose of defining *energy*. We will then show that energy becomes a more general concept in physics — beyond just the *kinetic energy* of moving bodies and the energy transferred from one object to another by *work done by a force* — to include the thermal physics concepts of heat, internal energy, and temperature, which are of primary concern in climate science. Specifically, it is considered a physical law of the universe that “energy is conserved”: although it might change from one form to another, every “drop” of it must be accounted for... it is neither created nor destroyed. The special case of *equilibrium*, where energy into a system is exactly balanced by energy out of a system, will be demonstrated using water as an analog for energy.

Note that there are a few tasks to complete within this section, highlighted with underlines.

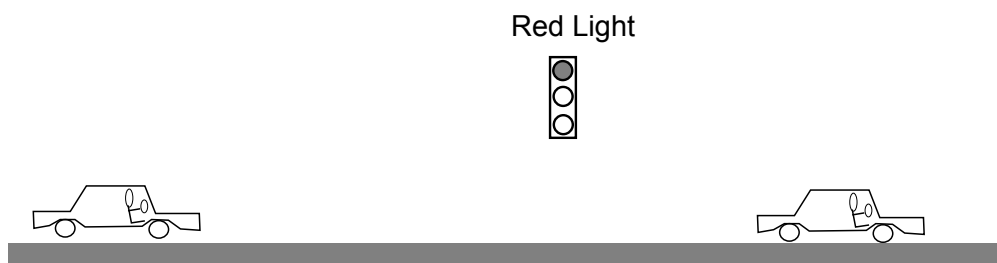
3.1.1 Motion

Before understanding *why* things move the way the do, we first must be able to *describe* motion. Physicists do this by defining three related quantities for an object:

$$\text{position: } x, \quad \text{velocity: } v = \frac{\Delta x}{\Delta t}, \quad \text{acceleration: } a = \frac{\Delta v}{\Delta t}$$

We will sometimes use vector notation for these quantities, e.g., $\vec{v}$ and $\vec{a}$, to show that they point in some direction (you don't just drive 65 mph, but you drive 65 mph East). But we will calculate using the above equations and assume that the motion is just along a straight line, called the x axis. In that case there are two “directions”: in the same direction as the positive x axis (positive values) and in the opposite direction (in the negative x direction, negative values).

Position is simply where the object is (e.g., on the x axis), and velocity should be mostly intuitive — *how fast is the object going* — but also contains the direction of motion. A nice way to understand the relationship between velocity and acceleration is to consider the cases of “speeding up” and “slowing down.” Consider the picture below and draw the pair acceleration and velocity vectors (arrows) for each car, where the left car is approaching a red light and the right one has just left the intersection after the light turned green. Label your vectors $\vec{v}$ and $\vec{a}$.



Now choose an x axis along the road by drawing an x - y axis (your *choices* are: the position of the origin, and the positive x direction). Then state whether each vector corresponds to a positive or negative one-dimensional value (you can just draw a “(+)” or “(−)” next to each).

3.1.2 Dynamics: Newton’s Law(s)

In 1687, Isaac Newton (English mathematician and physicist) published “The Principia¹,” which revolutionized human understanding of motion and its causes. Prior to Newton, it was believed that the physical law proposed by Aristotle (Greek, 384–322 BCE) explained motion, essentially: *Objects move when pushed, and stop moving when no longer pushed*². Following in the footsteps of Copernicus and Galileo, Newton proposed a better law of motion (**Newton’s Second Law**):

$$\text{Objects change their motion when pushed} \quad \text{or} \quad \sum \vec{F} = m\vec{a} \quad \text{or} \quad \vec{a} = \frac{\sum \vec{F}}{m}$$

where $\vec{F}$ is a **force** (e.g., a push or pull interaction), the notation $\sum \vec{F}$ means “the sum of all forces on the object”, and m is the **mass** of the object (e.g., in kilograms). So Newton discovered that interaction *forces cause changes in velocity*. Put another way: if there are no forces acting ($\sum \vec{F} = 0$) then there is no change to the velocity — the object continues to move with the same velocity (vector).

¹It was written in Latin and the full title was: *Philosophiæ Naturalis Principia Mathematica* (Mathematical Principles of Natural Philosophy).

²This might seem like a good law based on your own observations of the world (you’re not alone: it held for 2000 years!). But it turns out that it is simply a good law for motion in a frictional fluid, like air. If you add friction to Newton’s correct description of motion, you obtain Aristotle’s principle.

Go back up to the two accelerating car(s) at the stoplight. Draw a third vector for each car called “ $\vec{F}$ ” to indicate the force direction on the car that is causing its acceleration. What do you think is causing that force on the car(s)? (What is the *object* exerting the force?)

3.1.3 Energy: Mechanical Energy

The origin of the concept of energy in physics is through a quantity called **Work**:

Work is the energy transferred to an object by a force, found as *force times distance traveled*.

or, as an equation:

$$W = F\Delta x = \text{Force} \times \text{distance} \quad (\text{positive when } \vec{F} \parallel \Delta\vec{x}, \text{ negative when opposite})$$

The connection of work to motion is that work causes changes in velocity. Specifically, it changes what is called **kinetic energy** of an object:

$$K = \frac{1}{2}mv^2 \quad (\text{Kinetic Energy, motional energy}) \quad (3.1)$$

by the “**Work-Energy Theorem**”:

The total work on an object is equal to the change in its kinetic energy: $W = \Delta K$.

Another type of mechanical energy is **Potential Energy**, which is stored energy (the potential to do work and cause kinetic energy). Common examples are springs (compressing an object against a spring) and gravity (raising an object up); in each case, “releasing” the object will allow the stored energy to be converted into motional kinetic energy.

For mechanical systems that have no friction, energy is always *conserved*: it can change from kinetic to potential, but will not be lost or gained.

3.1.4 Energy: Other Energy

Materials constructed from atoms (either the free-flowing fluids of liquids and gases, or the rigid lattices of solids) contain **internal energy**. If we think of these systems as just collections of little particles, then: (i) a fluid’s internal energy is determined by sum of the kinetic energy of all particles bouncing around in the container; (ii) a solid’s internal energy is contained in the kinetic energy of the vibrating particles, and in the potential energy of the “springs” that connect them together. When the number of particles in a material becomes very large, we treat the object as a thermal system: the *temperature* is a measure of the average internal energy of the particles.

There are other types of energy, some of which we will consider later: energy from electromagnetic radiation (light), electrical currents, nuclear energy.

3.1.5 Energy: Equilibrium

For an object that is both receiving energy and giving away energy, we say that the object is in **equilibrium** when:

$$\text{Rate of Energy In} = \text{Rate of Energy Out} \quad (\text{Equilibrium})$$

3.2 Experiments and Activities

3.2.1 Motion

Find a “frictionless” track and car and bring it to your lab bench. Set it up and make sure that it is level by adjusting the “feet.” In each of these two experiments

1. On a level track, give the car a kick at one end, and have your partner stop the car before it reaches the other end.
2. Raise one end of the track and give the car a kick (not enough to go off the other end), then catch it.

perform the following tasks:

- Do the experiment with the car (be sure to catch it before it reaches the ends).
- Time the experiment to get a range of times for your experiment.
- Observe the motion of the car during the whole time.
- Plot a position (x) vs time graph, and a velocity (v) vs time graph below it, as we did in lecture.
- Identify on the graph any time periods where the car is *accelerating*. (How can you tell?)

3.2.2 Mechanical Work

Attach a pulley to the end of your track. Cut a long (3 m) piece of string and attach it to the car and to a mass-hanger with some amount of weight.

- Record the amount of mass on your mass hanger (including the hanger itself). Calculate the *weight*³ of this amount using $\text{weight} = F_g = mg$.
- Do a few trials where you release the mass hanger from just below the table, letting it pull the car.
- Construct the position and velocity graphs again for this motion.
- Describe how this experiment relates to *work* and the “*work energy theorem*”.

Return the track, the car, and the pulleys back to their places.

3.2.3 Work and thermal energy

Complete this with your whole table (4 people): one pair doing the stirring and one doing the hand-heating.

- At the sink, fill up a glass beaker with cold water.
- Then fill the two 50 mL graduated cylinders with the same water, to the same level (approximately 20 mL). Record the level and the temperature.
- One pair will attempt to increase the temperature of the water by *mechanical work* (stirring with the plastic rod).

³The Weight of an object is the force of gravity on the object, $W = F_g = mg$, where m is the mass in kilograms, g is the “acceleration due to gravity” (10 m/s^2). The units on this product, kilograms times meters per second squared, are called *Newtons*, N.

- The other pair will attempt to increase the temperature of the water by conduction with your hand.
- Conduct the experiment at least once for two minutes and record any temperature changes.

3.2.4 Equilibrium: Leaky Buckets

Over the next few weeks in class, we will consider the energy equilibrium of the Earth, meaning the balance between the incoming energy per second from the Sun and the outgoing energy per second that the Earth emits into space. It turns out that, while the incoming rate (energy from the sun) is approximately unchanging, the outgoing rate *depends on the Earth's temperature*. As an equation, it looks like this:

$$\text{Rate}_{\text{Sun Energy In}} = \text{Rate}_{\text{Earth Energy Out}}(T_{\text{Earth}})$$

where the parentheses on the right-hand side means that the rate *depends* on temperature. The importance of this dependence is that in a situation of equilibrium (when the left-hand side equals the right-hand side) the result implies a single temperature: we can determine the temperature of the Earth by assuming equilibrium.

Today we will play with an analogy of this situation: a “leaky bucket” that is being filled by a variable-rate pitcher. We will see if the equilibrium flow rate depends on any properties of the system (like the rate of energy into/out-of the Earth depends on its temperature).

Complete the following, using the data table on the next page, and eventually making a graph on graph paper of the last two columns.

- Fill the pitcher of water about 3/4 full.
- Raise the blue “lab jack” up so that the 100 mL graduated cylinder will fit underneath the pitcher spout.
- Place the catch tray below the spout and then connect the aquarium pump: the pump will be sitting on the table with one rubber tube (IN) going into the catch tray, and another rubber tube (OUT) going into the pitcher. You can test that this is working by plugging in the pump and letting the water flow from the pitcher into the catch tray, and back to the pitcher through the pump.
- Place the 100 mL graduated cylinder (GC) in the catch tray and let the water go out of the spout into the cylinder. Record any general observations.
- Now adjust the flow from the spout so that the level of the water in the GC is just a little bit above the hole (couple mL).
- On the table, record the *baseline* level (the value on the GC of the center of the hole).
- In “Approx Level” put “just above hole”, and record its “Exact Level”, and then subtract off the baseline value to get the ΔV value.
- Now remove the GC without changing the flow.
- Use the 25 mL graduated cylinder and the stopwatch to time how long it takes to fill (to the 25 mL line). Record this in your table under “Time to 25 mL”. It would be best to do this three times, take the average and record the average and approximate uncertainty.

You can then calculate the “Flow Rate” (or you can wait to do that later for all of the data you collect).

- In the rest of the rows of the table, choose successively larger flow rates. The “Approx level” could give you a target to try and find (like “twice the ΔV as the first”, or three times, etc), but always record the exact level that you have.
- In the end calculate the height in cm from the ΔV column by measuring the distance on the GC itself.
- On graph paper, make a plot of your data: either Flow rate vs Height, or vice-versa⁴.

⁴When we say “y vs x” we mean “vertical vs horizontal”.

Approx level	Exact Level	ΔV (above hole)	Time to 25 mL	Flow Rate	Height
()	()	()	()	()	()
Baseline Value (Hole Center)					

Be sure to fill in the units in the second row.

Lab 4

Electromagnetic Radiation and Blackbody Radiation

All the light you cannot see

Equipment List

- IR camera (w/ cable)
- NIR security camera (w/ cable)
- “go-pro” camera (w/ cable)
- NIR LED light panel
- wooden block (“stage”)
- bathtoy (“character”)
- 250 mL beaker
- cardboard box
- large (12in square) plastic window
- small (3in square) plastic window
- small plastic window, blacked out
- TV remote
- Silicon disks
- Coke or Diet Coke or Coke Zero
- Faraday Cage with radio

Note: Today will make observations of light with a few different cameras. You will submit on Blackboard the usual pdf of the lab assignment and your written work, but then you can just submit extra images (jpg files) in addition the pdf (**do not print images or add them to your pdf**). If you need to correct a submitted pdf, just submit another one; I will only use the last pdf document. Image files can be obtained from the IR thermal camera by plugging it into the computer with the USB-C to USB cable; it will appear as a drive with a DCIM directory. Images from the “go-pro/active” visual light camera and the near-IR “security” webcam can simply be screenshotted or downloaded using the Camera app on the windows PC. **Every student should get their own copies of the relevant images** and save them to their email or to, e.g., drive.google.com.

4.1 Introduction and Definitions

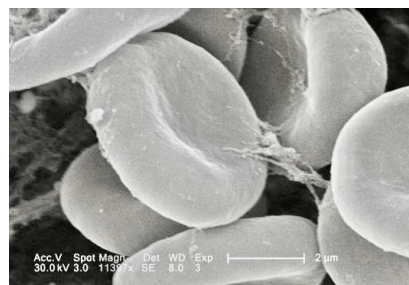
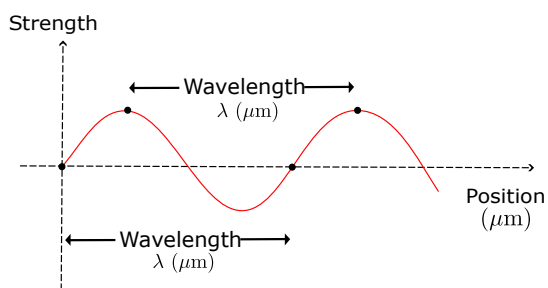
Today we consider light, in all of its colors/wavelengths. The collection of light of all wavelengths is called the **electromagnetic spectrum** (because light, it turns out is a propagating wave of

electric and magnetic fields). Objects with a nonzero absolute temperature (having any internal energy) will **radiate** light of a wavelength/color that is specific to their temperature. The idealization of this is called **blackbody radiation**, because perfect emitters are also perfect absorbers (“black”).

4.1.1 Electromagnetic Spectrum

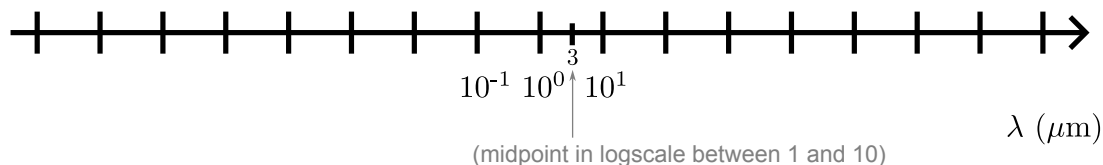
Light is an oscillating **electromagnetic wave** with a characteristic wavelength, which we usually write as the greek letter “lambda,” λ . The range of wavelengths that we are able to see with our eyes is called **visible** light and spans a range of wavelengths from $0.40\ \mu\text{m}$ (blueish light, $400\ \text{nm}$) to $0.75\ \mu\text{m}$ (reddish light, $750\ \text{nm}$).

Consider the schematic wave below and the red blood cell given for scale. Sketch two waves, one that has the wavelength of red light, and one that has the wavelength of blue light, making them to scale with the blood cell image.



Now consider the rest of the electromagnetic spectrum. We give the names to ranges of wavelength values: gamma-rays, X-rays, ultraviolet, visible, near-infrared, infrared/far-infrared, microwave, radio.

- Look up the wavelength ranges of these types of light and make a table, putting all wavelengths in micrometer (μm).
- On the wavelength axis below (in log scale) show the range of each type of light. [Notice that $10^0\ \mu\text{m} = 1\ \mu\text{m}$ (and $10^1 = 10$, $10^{-1} = 1/10$), and that the midpoint between 1 and 10 on a logarithmic scale is 3.]



4.1.2 Blackbody Radiation

Any object that has internal energy will have a temperature greater than absolute zero (0 K = -273°C); and it turns out that any object with a temperature radiates light. This phenomena is called **blackbody radiation**. There are a few key equations that describe the amount and wavelength of light emitted by blackbody radiators:

$$\lambda_{\text{peak}} = \frac{2900 \mu\text{m} \cdot \text{K}}{T} \quad \text{Wien's Law, wavelength of peak emission}$$

$$F = \left(6 \times 10^{-8} \frac{\text{J/s}}{\text{m}^2 \cdot \text{K}^4} \right) T^4 \quad \text{Stefan-Boltzmann Law (total flux)}$$

$$F_{\lambda} = \frac{F}{1.5 \times \lambda_{\text{peak}}} \quad \text{Peak value of "Spectral Flux"}$$

Our Sun is the most familiar example of a (nearly) ideal blackbody radiator. It has a temperature of about 6000 K. What is the wavelength of the peak of its blackbody radiation? What is its total flux?

A table is given below showing the spectral emission or sensitivities of equipment we will use in the lab today. Fill in the last column with the temperatures corresponding to blackbody radiators with peaks at the wavelengths given (one temperature if a single value, two temperatures if a range is given). There is space for calculation examples below.

Device	Wavelength range (μm)	BB Temperatures
Mileseey IR Camera	4.6 μm – 11.5 μm	
ELP USB NIR (webcam)	0.4 μm – 1.0 μm	
ELP USB NIR (illuminators)	0.94 μm	
Andoer IR49S LED Panel	0.855 μm	
Andoer IR49S LED Panel	0.855 μm	
AKASO v50 elite action camera	0.40 μm – 0.70 μm	

4.2 Experiments and Activities

4.2.1 Observations of light

We have equipment for observing light in the (far) infrared (the Mileseey camera); in the near infrared (NIR) (the ELP webcam); and the visual range (your eyes and the AKASO “action camera”). For the NIR (and comparisons between NIR and visual) you can put the cameras and objects under the cardboard box to get a dark environment. Objects with a temperature (at your table, around the room) can serve as sources of infrared light. The Andoer LED panel acts as a source of near IR light.

For each item of observation, make and document (with words and images) some observations. Some things to try, and questions to answer:

- Use the infrared camera to observe objects and people (with their permission!) around the room.
- Try to see the “ghosts” of warm objects when they are removed from another object (your hand on the table or wall).
- Spill a bit of water on the floor and see what it looks like.
- See if various lights change in the IR picture when on/off.
- What are warm objects and cold objects in the room ... anything surprising? (Note the temperature output on the image itself).
- Place the security camera and the visual camera under the box with a “stage” (little block) and a “performer” (bath toy animal). How do their view compare?
- Put the NIR LED panel in the box as well, and document the visual and NIR camera views.

- Can you see the NIR LEDs with your eye?
- Can anything see the bulb in the TV remote when you press its buttons?
- How does the NIR webcam see things? Is it the same idea as the IR camera? How do such “security cameras” work, and why are they designed this way?

4.2.2 Observations of transmission of light

Whether an object is called **transparent** or **opaque** turns out to be wavelength dependent. What does this mean in plain language?

Now make some observations. Some things to try and questions to answer:

- Use the “clear” plastic window, with various sources and cameras.
- Use the silicon plates (be careful they are fragile, but you can put your fingers on them) with various sources and cameras.
- Try the glass in the doorway.
- Fill a small beaker with coke. Place an object (pencil?) in it and observe with your eyes, the action camera, and under the box with the NIR camera and/or the visible light camera, with and without the NIR LED panel.
- Try out the “faraday cage” and the AM/FM radio. Two local stations are WGVU (88.5 MHz) and WOOD (1300 KHz). What are the wavelengths of these broadcast stations (calculate using $c = \lambda f$, where the speed of light is $c = 3 \times 10^8$ m/s).
- Does the faraday cage affect cell-phone signals?

Lab 5

Heating by Radiation

Feel sun on your face. No one else can feel it for you.

Equipment List

- Calorimetry kit: Calorimetry can, aluminum cylinder, 250 mL beaker
- Heating by Radiation kit: aluminum block, styrofoam holder with kickstand, no-slip pad, large mass
- Digital thermometer
- 100 mL graduated cylinder
- styrofoam cups (for getting ice)
- incandescent desk lamp
- graph paper
- cooler of ice (for room)
- scale (for room)
- active camera (optional)
- computer (optional)

5.1 Introduction and Definitions

To estimate the energy balance of the Earth, and thus estimate its expected equilibrium temperature, we must first know the input energy to Earth system: the radiation flux (in $\frac{J/s}{m^2}$) from the Sun. Recall our “leaky bucket” experiment: When you increased the flow of water coming out of the pitcher, you observed a rising water level in the graduated cylinder (the leaky bucket), until the point at which the rate of water spraying out of the hole match the incoming rate of flow from the pitcher. When that equilibrium was achieved, there was an equilibrium height of the water above the hole¹. We can make an exact analogy to the energy balance of the earth: the Sun pours energy into the Earth through photons of light; the Earth sprays energy back into space because it has a temperature (specifying its internal energy) and therefore radiates as a blackbody; and when these two rates of flux are equal, the internal energy of the Earth stops changing, leading to an “equilibrium level” of temperature. Today we would like to directly measure the amount of energy being “poured in” by the Sun. (Or, if the Sun does not cooperate, an incandescent light bulb).

¹That height determines the rate of flow out of hole, essentially by conservation of mechanical energy: a larger column of water in gravity, leads to a larger kinetic energy of the exiting water.

5.1.1 First Law of Thermodynamics, Heating, and Calorimetry

In our introduction to the topic of energy — beginning with the work by a force, generalized to include all types energy transfer mechanisms, and all types of internal and external energy types — we emphasized the basic physical law: energy is conserved². Most generally, energy conservation is stated as a law within thermodynamics (the physical topic of heat, temperature, and chemistry) and is given the grand title **The First Law of Thermodynamics** (FLTD):

$$\Delta E_{\text{tot}} = Q + W.$$

This equation pertains to any single object/system that you choose and summarizes the rules we used in the construction of energy flow diagrams: a change in the total energy of an object, ΔE_{tot} , must be exactly accounted for by the energy input to or output from that object (on the right hand side), in the form of **heat flow**, Q , or mechanical **work**, W (these flows, Q and W , are positive when energy is coming into the system, and negative when energy is going out).

Today we will consider a special case of the above equation, where our system consists of two objects brought together at different temperatures. There will be no work done on the system and we will consider the combined system of two objects to be “closed” so that there is no heat flow in or out of the system. In that case, we have:

$$\Delta E_{\text{tot}} = Q + W \rightarrow \Delta E_{\text{tot}} = 0 + 0 \rightarrow \Delta E_1 + \Delta E_2 = 0,$$

which says that the sum of the internal energy changes in the two sub-objects must equal zero. Another way to view the combined system is to break it into two objects. In that case we would have two equations (one equation for each “object” box, in an energy flow diagram):

$$\begin{array}{lcl} \Delta E_{\text{tot},1} = Q_1 + W_1 & \rightarrow & \Delta E_{\text{tot},1} = Q_1 + 0 \\ \Delta E_{\text{tot},2} = Q_2 + W_2 & \rightarrow & \Delta E_{\text{tot},2} = Q_2 + 0 \end{array} \rightarrow Q_1 + Q_2 = 0$$

where the last equation follows from the previous two, plus our prior equation for the combined system (that the sum of the two energy changes is zero). When drawing an energy flow diagram, there is a single transfer arrow between the two objects, which we could label with $|Q|$. In the equation for one object, that transfer is accounted as a positive or negative value depending on whether the flow is into or out of that object.

Consider an example system: cold milk poured into a mug of hot coffee. We will ignore the energy loss out of the system as a whole during the time of mixing, so the above equation holds. Sketch an energy flow diagram for these two objects during the time they are mixing (not yet at equilibrium). Then write down three FLTD equations, like those above: one for the combined system and one each for the objects in your diagram. Instead of “1” and “2” labels, use “C” and “M”.

We express heat flow into an object through an equation that demonstrates its change in the internal energy of that object (essentially The **heat capacity** of an object, C , is the constant of proportionality between the heat transfer to/from the object and its internal energy change, as measured by a change in temperature:

$$Q = C \Delta T.$$

²Although it may change its form, and transfer from one object to another, the total amount of energy never changes.

This is not a very useful equation because the value of C depends on all of the details of the object. A more fundamental quantity is the **specific heat capacity** (often called simply the “specific heat”) of a *material*, denoted by the lower-case c . This is the heat capacity per mass of an object, so we can write:

$$Q = c m \Delta T.$$

One of the materials with the largest specific heat capacity is water, with $c_W = 4.2 \text{ J/g/K}$. This means that it takes 4.2 J of energy to raise one gram of water one degree Kelvin (which, since $\Delta T_K = \Delta T_{\circ C}$, is the same amount to raise by one degree Celsius). The term “capacity” is used because, once you have added energy to raise the temperature, that energy is essentially stored in the water and can be transferred somewhere else. Other materials that have a lower specific heat capacity, like aluminum, experience a higher temperature increase for the same, 4.2 J, energy input — or, put another way, raising the temperature of aluminum by one degree stores less energy than raising the same amount of water by one degree (aluminum’s capacity to store energy is less than water).

Write down a $Q_1 + Q_2 = 0$ equation for the “milk plus coffee” example above. Again, use “C” and “M” as labels. Then use the above equations containing specific heat to express heat as a temperature difference. The final temperature within the two ΔT ’s will be the same, T_f , but the initial temperatures will be different. Re-write your equation to show this in variables. Then discuss briefly why the two terms will add to zero, knowing what you do about the initial and final temperatures of the coffee and milk.

5.2 Experiments and Activities

5.2.1 Specific heat of aluminum

Before looking at the heating of an aluminum block under the radiation of the sun (or incandescent bulb), we will first determine the specific heat capacity of the material itself. Our experiment will be a very simple calorimetry example: prepare a small aluminum cylinder at a known temperature (letting it equilibrate with ice water) and then drop it into a vat of water while monitoring the temperature of the water. The drop in water temperature at that instant can be associated to a heat transfer from water to aluminum, which causes an internal energy decrease/increase, respectively. Knowing the specific heat water (and the masses of both the water and aluminum), we can determine the specific heat of aluminum.

Procedure: Using your own paper to write down observations, keep a data table, and make calculations from equations, take the following basic steps.

- Open up the “calorimetry can” and inspect it. What are the components of it, and what is the purpose of each? (The small cup inside is what you will fill with water).
- Get some ice from the cooler. Create an ice-water bath in your 250 mL beaker and place the aluminum cylinder inside.
- Get hot water from the faucet (50°C) and fill the small calorimetry cup with about 2 cm (we will measure this amount after the experiment is over).
- Place the cup inside the calorimetry can and observe the temperature through the hole in the cork.
- Start the stopwatch and take temperature measurements for at least 5 minutes, at approximately 30 second intervals³.
- While you are still getting temperature measurements on the water, use a second thermometer, to check that the ice water and cylinder mixture is at the right temperature.
- Remove the cylinder using the tongs and drop it into the warm water.
- Record the time that the cylinder was added.
- Continue to measure the temperature for another 5 minutes, at 30 second intervals.
- Create a table of your temperature vs. time data. It is probably easiest to simply record the time on the stopwatch (4m50s) and then later you can convert that (in an additional column) to the number of seconds (290 s).
- Make a graph of temperature vs. time on graph paper. Be sure to choose axes such that your data covers most of the page.
- Using a ruler, make two best-fit straight lines, before and after the time at which the cylinder was inserted.
- At the time the cylinder was inserted, estimate the vertical distance (temperature distance) between these two lines. Also record the “final” temperature (the bottom of this range).
- Carefully remove the aluminum cylinder from the calorimetry can (without losing any water).

³If you would like, you could use the go-pro-like camera to record the stopwatch and thermometer — probably best to record with the “time-lapse” feature — and then get (temperature, time) data points by watching the video afterwards. Think carefully about how to do this ahead of time.

Then, using the digital scale, measure and record the mass of the aluminum cylinder.

- Take your small calorimetry cup to the sink, along with your 100 mL graduated cylinder. Pour the water into the graduated cylinder to record the volume of water in the can. (Think ahead of time about how to do this if your water turns out to be more than 100 mL in volume!)
- Draw an energy flow diagram that will captures the heat flow and energy changes during the time immediately after placing the aluminum cylinder in hot water, but before it equilibrated.
- Then write down equations to arrive at a $Q_1 + Q_2 = 0$ equation, like you did in the introduction. Use this final equation to calculate the specific heat capacity of aluminum. Recall that the specific heat capacity of water is $c_W = 4.2 \text{ J/g/}^\circ\text{C}$.

5.2.2 Heating aluminum block by radiation

Now that we know the specific heat of aluminum, we can measure precisely how incoming radiation flux causes a temperature increase. In this experiment the heating of an aluminum block will be accomplished by light (from the sun or your incandescent desk lamp) rather than conduction from a hot to a cold object. But the input heat for aluminum can be interpreted with exactly the same equation:

$$Q_{\text{Al}} = m_{\text{Al}} c_{\text{Al}} \Delta T$$

Procedure: Using your own paper to write down observations, keep a data table, and make calculations from equations, take the following basic steps.

- Place your aluminum block into the insulating styrofoam. If there is no hole through the styrofoam for the thermometer, measure the position of the hole in the block and carefully figure out where to pierce the styrofoam such that the thermometer will hit the hole.
- Prop up the styrofoam at an angle, using its kickstand, the no-slip pad and a large (e.g., 1 kg) mass.
- Arrange the desk lamp so that it is pointing at the block and quite close ($\sim 5 \text{ cm}$). Record the distance.
- Start your stopwatch.
- Then turn on your desk lamp.
- Record (time, Temperature) values in a table, every 30 seconds (or even every minute should be fine) for about 20 minutes.
- Make a graph of your data, Temperature vs. time.
- Using a ruler, draw a straight line of best fit. Then, using two widely-separated *points on the line* calculate the slope of that line.
- This slope, $\frac{\Delta T}{\Delta t}$, can be converted into a $\frac{\Delta E}{\Delta t}$ using the specific heat capacity of Aluminum. Write this out as an equation and then calculate the value.
- The power increase you just calculated can be translated into a flux, by dividing by the area of the aluminum block. Write out an equation for this, and then calculate a value.
- Using your distance to the lamp, you can now try to estimate the power of the light bulb. This is the same procedure we did in class to go from the Sun's power, to its flux at the position of the Earth's orbit, but backwards.